Group Assignment

TECHNOLOGY PARK MALAYSIA

AICT020-4-2-EDAV

Exploratory Data Analytics & Visualization

UCDF2407ICT(DI)

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EDAV REPORT

Dataset Name: Retail Clothing

Summary of the Cleaned Dataset

The data transformation and preparation process followed a strict, auditable sequence to establish a clean and robust dimensional model. The process began with a preliminary audit of all imported tables to verify the initial data types inferred by the system. Subsequently, all original source tables (e.g., "Customer," "Sales") were duplicated, and the working copies were appended with the (Clean) suffix (e.g., "Customer(Clean)," "Sales(Clean)") to ensure a non-destructive data cleaning workflow (Databricks, 2023). All subsequent transformations were applied exclusively to these copies, including trimming all text columns for uniformity, merging customer name fields into a single Customer Name column, and deriving the Age from the birth date field to enable demographic analysis. After cleaning, the final star schema architecture was implemented by referencing these processed tables: "Customer(Clean)" became the DimCustomer table, "Product(Clean)" became DimProduct, and "Sales(Clean)" became the central FactSales table. A dedicated DimDate table was also created to support time intelligence calculations. Finally, all dimension tables (DimCustomer, DimProduct, DimDate, etc.) were connected to the central FactSales table via one-to-many relationships, structuring the dataset into a scalable and efficient star schema optimized for business intelligence reporting (Tobin, 2025).

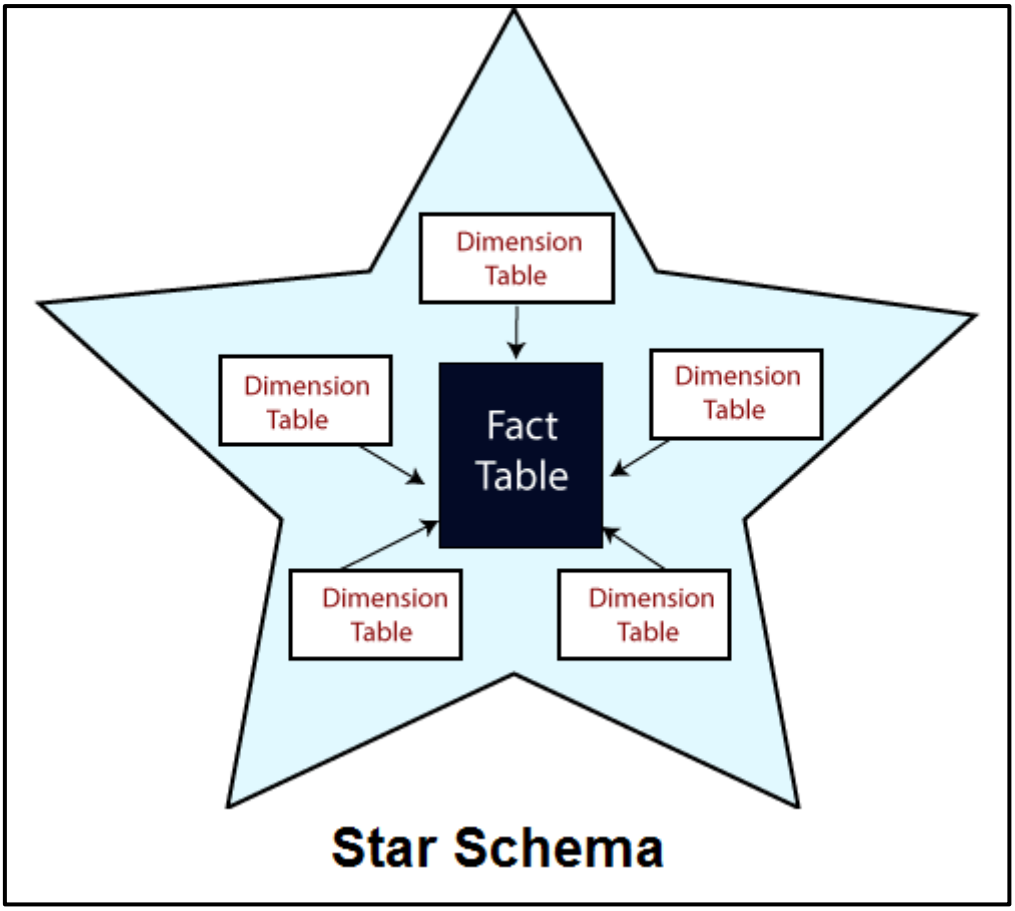

Figure 1: Star Schema

This rigorous data preparation phase is fundamentally necessary because raw data is rarely fit for direct analysis. Real-world datasets invariably contain inconsistencies, duplicate entries, incorrect data types (such as text stored as numbers), and missing values. Without thorough cleaning and transformation, these imperfections introduce bias and inaccuracy into the analytical results and measures, leading to unreliable business insights and flawed decision-making. By implementing robust cleaning steps—such as establishing proper relationships, ensuring data quality, and structuring the data into a Star Schema—the dataset becomes a single source of truth that provides the foundation for accurate calculations (like AOV and retention rates) and reliable visualization, thereby fulfilling the core purpose of exploratory data analysis *(Maurya, 2025)*.

Explanation of Each Visualization & Statistical Findings

Global Interactive Controls and Unified Filter Interface

The Team BMW report and dashboard are designed as an interactive report rather than a static reporting sheet. All information and controls are in the Global Filter Interface, which serves as the Command Centre of the entire report suite. By placing these controls prominently, users will quickly identify how they can move from high-level reports to a very detailed understanding of operational activities with one click.

Unlike standard filters that are typically found in a side panel, slicers that are displayed directly on the canvas allow the user to see the active state of the data immediately. Whether it is a Core Performance or Category Deep-Dive report, the slicers will create a single filter context for all visuals. Therefore, any choice of "Year" or "Region" will be applied to all visualisations, ensuring a single truth for analysis and preventing an all-too-common mistake, comparing two data sets that are not aligned.

In addition, this design focuses on providing a high level of user autonomy. It allows management to respond to their own follow-up questions without needing to wait for a new report, therefore greatly increasing the efficiency and speed of data-driven decision-making.

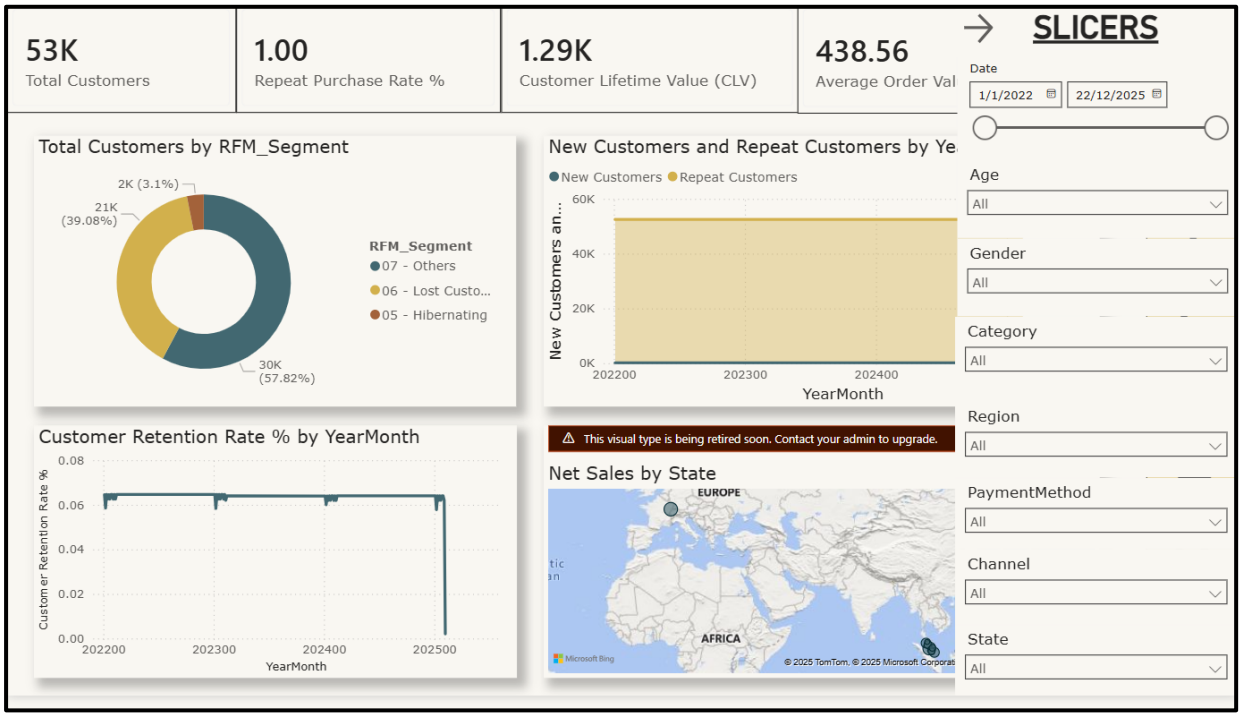

Figure 2: Global Report Slicers

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Statistical Finding

Group 1: Volume & Profitability

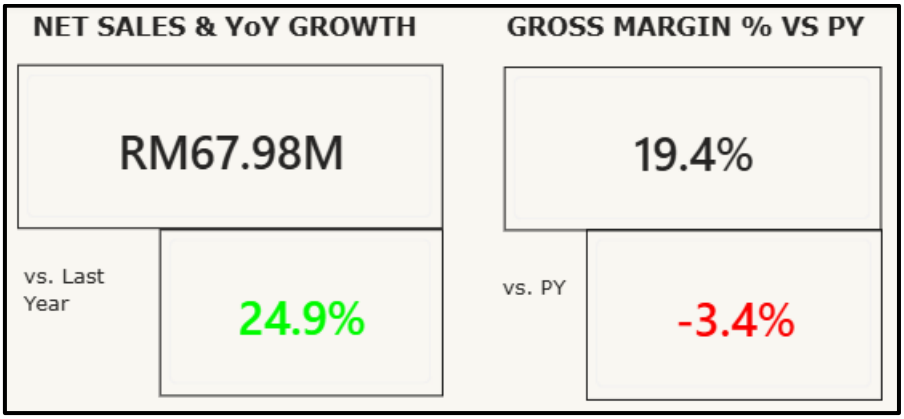

Figure 3: Volume & Profitability KPI cards

Element	Description
KPIs Explained	**Net Sales & YoY Growth** and **Gross Margin & PY**.
Purpose	To begin assessing the health of the business and identify the key financial issue(s) it has, which can be boiled down to the relationship between the volume of sales achieved and the profitability of those sales. The subsequent diagnostic visuals will be based on this relationship.
Statistical Findings	The amount received for net sales is RM67.98 million, which equates to a substantial positive year-over-year increase of 24.9%. On the other hand, the total gross profit margin percentage is currently 19.4% and has deteriorated 3.4% compared to the previous year.
Narrative Interpretation	The results suggest a puzzling situation. A high volume of sales, with a 24.9% increase, means strong market demand and/or the effectiveness of the sales strategy. However, the business's ability to generate profit has been negatively affected by the same sales volume decline. Therefore, the immediate area of concern for the business revolves around internal operations and/or increasing costs associated with the sales (e.g., cost of goods sold, discounting, and/or pricing strategy) rather than generating revenue.

Group 2: Efficiency & Risk

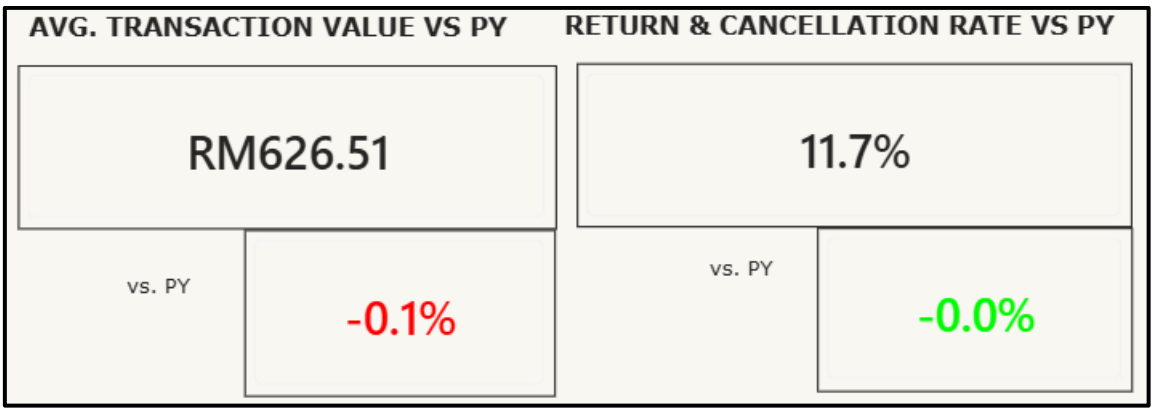

Figure 4: Efficiency & Risk KPI cards

Element	Description
KPIs Explained	**Avg. Transaction Value (ATV)** and **Return & Cancellation Rate**.
Purpose	To understand how Group 1 defined the type of decrease in efficiency they are experiencing, these performance metrics are used to ascertain whether the margin issue is related to diminishing basket sizes (ATV) or product dissatisfaction (Return Rate).
Statistical Findings	In comparison to the previous year's average transaction value of RM626.51, it remains at a similar level with a marginal decline of -0.1%. The return and cancellation rate remains constant at the same rate of 11.7% with no deviation from the previous year.
Narrative Interpretation	The stable Return and Cancellation Rate (-0.0%) indicates that the reason for declining primary Gross Margin isn't likely to be attributed to poor quality of product or customer dissatisfaction due to returns. Additionally, the small but persistent decline of Average Transaction Value (-0.1%) indicates that there continues to be a challenge with maximizing the value of each order (e.g., failed upsell and/or cross-sell initiatives). This decline also negatively impacts the overall profitability trend, as noted in Group 1.

Visual 1: Line and Stacked Column Chart – Sales & Profitability Trend

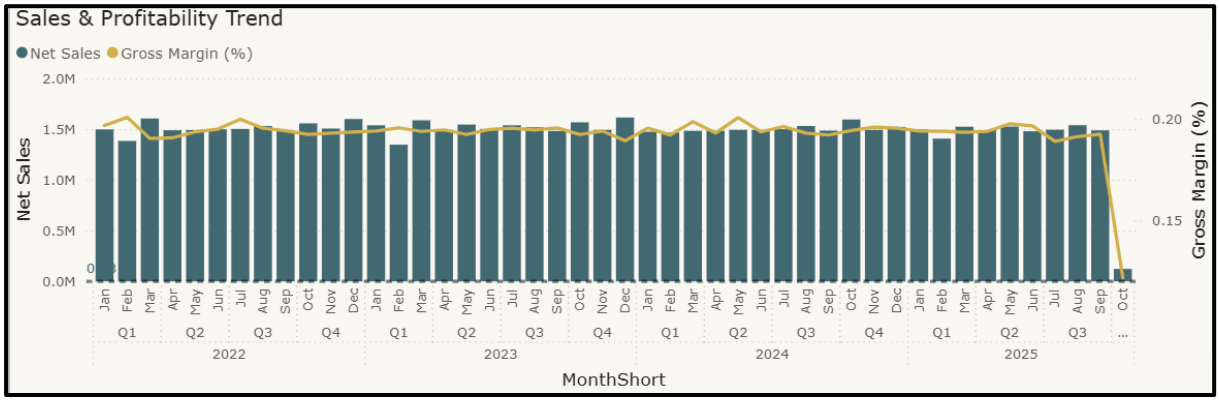

Figure 5: Sales & Profitability Chart

Element	Description
Purpose	A longitudinal diagnostic view that provides information on when the trends in Net Sales (Volume) diverge from the trends in Gross Margin (%). The design allows for quick and convenient assessment and diagnosis of core business issues that result from the difficulty in comparing metrics that have different scales (i.e., one is in value terms, while the other is in percentage terms).
Visual Type	**Line and Stacked Column Chart**
Fields	**Columns (Primary Y-Axis):** Net Sales **Line (Secondary X-Axis):** Gross Margin (%) **X-Axis:** Year Hierarchy (Time Series: Year, Quarter, Month)
Interpretation	By separating the volume (shown in the columns) from efficiency (represented by the line), the chart shows that the Net Sales area has demonstrated that sustained market demand is evident through a steady and/or increasing trend in sales volumes, as noted by a total increase KPI % of +24.9. The concurrent decrease and/or variance in the Gross Margin line indicates that the company's challenge is less about market demand and, instead, internal efficiency (i.e., rising costs, pricing failures). This method of using a chart with two axes is the only way to separate the success of volume achieved from the failure of profitability obtained.

Visual 2: Clustered Column Chart – Average Order Value by Sales Channel

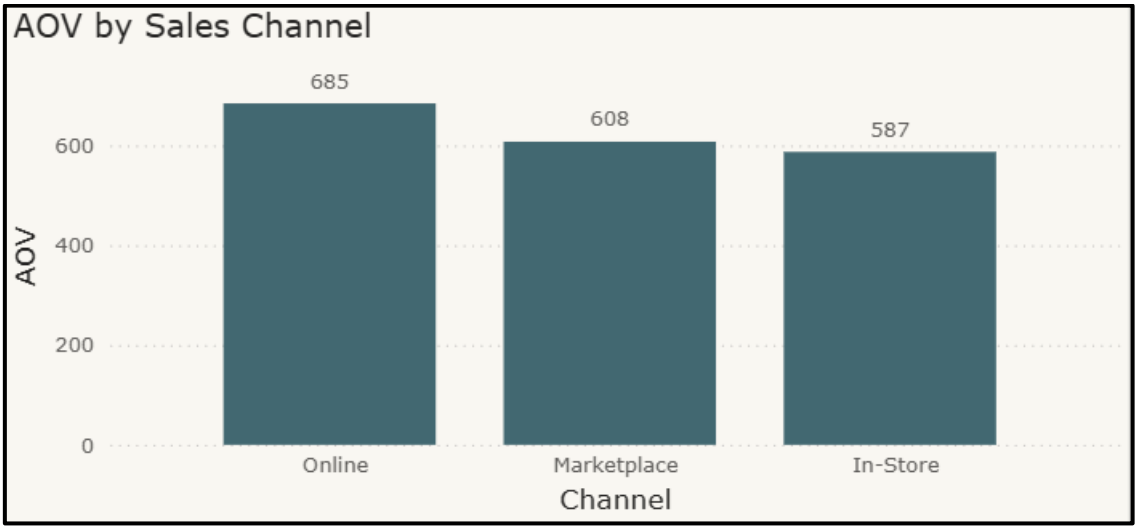

Figure 6: AOV by Sales Channel Chart

Element	Description
Purpose	To identify which sales channels are driving the highest and lowest Average Order Value (AOV). This diagnosis is crucial for improving the overall profitability trend, as maximizing AOV directly counters the slight decline noted in the KPI cards (−0.1%).
Visual Type	**Clustered Column Chart**
Fields	**X-Axis:** Sales Channel (Online, Marketplace, In-Store) **Y-Axis:** AOV (Average Order Value)
Interpretation	The visual data clearly illustrates that the performance level for each channel differs greatly between the various Channels. The Online Channel has the highest average order value (RM685). This indicates that customers tend to bundle or purchase larger quantities of items most readily online as opposed to in-store. Whereas, the In-Store Channel shows customers purchasing the lowest average order values (RM587). Therefore, there appears to be an opportunity for the implementation of strategies related to further enhancing Visual Merchandising, Targeted Impulse Purchases, as well as improving training for Sales Associates in order to create larger baskets at the Store's Check-Out Counter.

Visual 3: Clustered Bar Chart - Margin % by Product Category

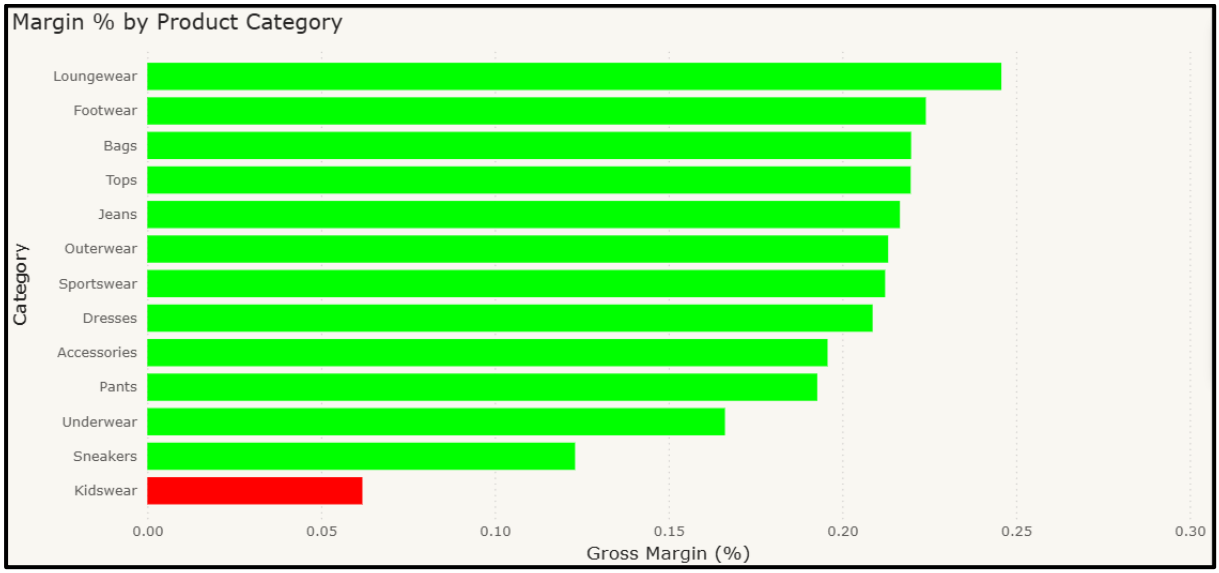

Figure 7: Margin & by Product Category Chart

Element	Description
Purpose	To identify which sales channels are driving the highest and lowest Average Order Value (AOV). This diagnosis is crucial for improving the overall profitability trend, as maximizing AOV directly counters the slight decline noted in the KPI cards (−0.1%).
Visual Type	**Clustered Column Chart**
Fields	**X-Axis:** Sales Channel (Online, Marketplace, In-Store) **Y-Axis:** AOV (Average Order Value)
Interpretation	The visual data clearly illustrates that the performance level for each channel differs greatly between the various Channels. The Online Channel has the highest average order value (RM685). This indicates that customers tend to bundle or purchase larger quantities of items most readily online as opposed to in-store. Whereas, the In-Store Channel shows customers purchasing the lowest average order values (RM587). Therefore, there appears to be an opportunity for the implementation of strategies related to further enhancing Visual Merchandising, Targeted Impulse Purchases, as well as improving training for Sales Associates in order to create larger baskets at the Store's Check-Out Counter.

Visual 4: Donut Chart – Revenue Share by Product Category

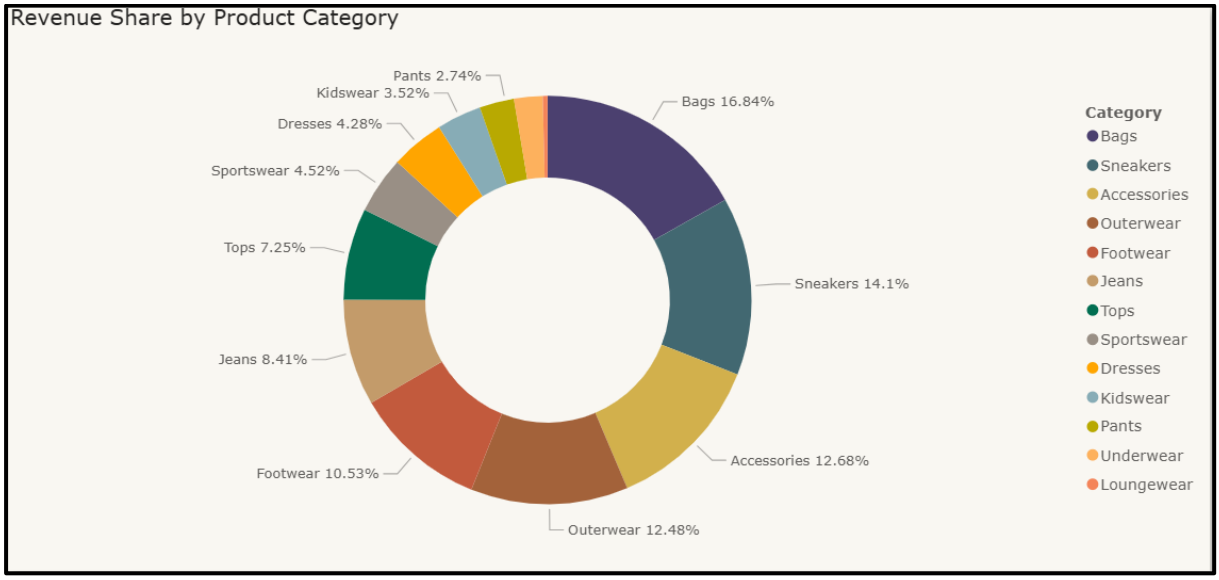

Figure 8: Revenue Share by Product Category Chart

Element	Description
Purpose	Product category proportions by total revenue. The data visually shows management categories of "Heroes" responsible for most of the business while also identifying how much those products will affect other categories that are performing poorly, as shown in other visuals.
Visual Type	**Donut Chart**
Fields	**Legend:** Product Category **Values:** Percentage share of total revenue
Interpretation	The data presented in the graph depicts Bags as having the highest contribution towards total sales at 16.84%. Sneakers are next, contributing 14.10%. In conjunction with the margin report, physical evidence that these categories generate both high volume and provide healthy margins exists. Kidswear has a representational total of only 3.52% of total sales, although the margin on Kidswear is very low compared to others in the kids' apparel category; therefore, this would not be considered a systemic risk for overall company profitability, but there does exist an opportunity to increase operational efficiencies in this product category.

Visual 5: Matrix Table – Detailed Margin & Sales Breakdown

Detailed Margin & Sales Breakdown

Category	Gross Margin (%)	Net Sales
Loungewear	0.25	255,936.34
Footwear	0.22	7,160,563.02
Bags	0.22	11,444,897.60
Tops	0.22	4,927,239.26
Jeans	0.22	5,714,674.28
Outerwear	0.21	8,486,463.64
Sportswear	0.21	3,075,774.38
Dresses	0.21	2,908,831.66
Accessories	0.20	8,617,124.67
Pants	0.19	1,864,426.78
Underwear	0.17	1,545,102.54
Sneakers	0.12	9,582,020.46
Kidswear	0.06	2,393,127.93
Total	**0.19**	**67,976,182.56**

Figure 9: Detailed Margin & Sales Breakdown Table

Element	**Description**
Purpose	A detailed tabular representation of product categories' performance metrics for granular analysis. This feature allows you to match Gross Margin % and Net Sales (Volume) to identify where the business has generated profits, as well as where efficiencies are missing.
Visual Type	**Matrix Table**
Fields	**Rows:** Product Category **Values:** Gross Margin (%) and Net Sales
Interpretation	The table provides final conclusive evidence by illustrating the numerical foundation of the visual representation. While the company achieved a net sales figure of RM67,976,182.56 with 19% overall margin, it indicates an obvious gulf in efficiency. The significant contributor is Bags, as a strong performer, being the number one-selling category, generating RM11,444,897.60 in sales with a high 22% margin. By contrast, Kidswear is specifically referenced as the major contributor to inefficiencies with only a 6% margin (in red), although it generated over RM2.39M in sales. With this data breakdown, stakeholders can transition from visual observations to more tactical decisions regarding categories.

Visual 6: 100% Stacked Column Chart – Net Sales Composition by Month

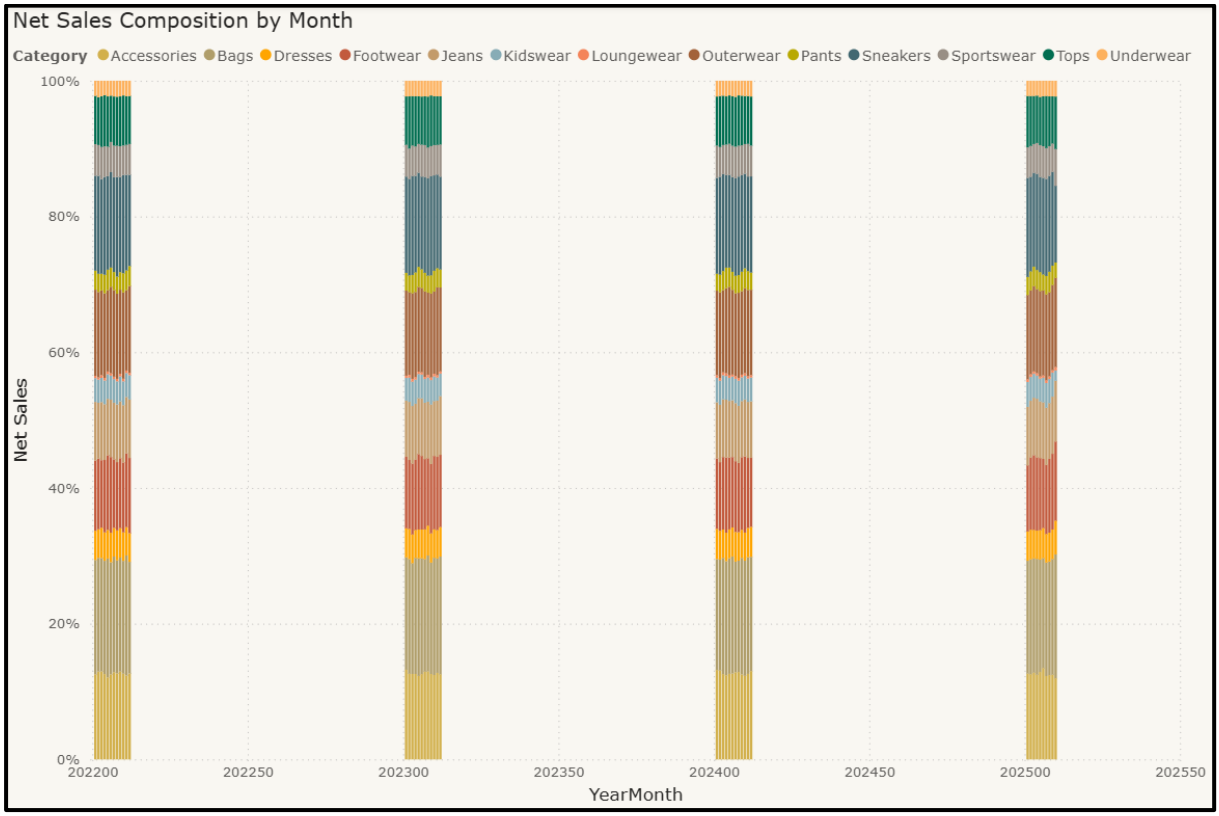

Figure 10: Net Sales Composition by Month Chart

Element	Description
Purpose	The analysis will be used to assess how much different product categories are contributing to total sales over an extended period of time and whether changes in total profitability are due to consumers shifting their purchasing patterns (for example, increased purchasing of low-margin products).
Visual Type	**100% Stacked Column Chart**
Fields	**X-Axis:** YearMonth **Y-Axis:** NetSales **Legend:** Product Categories
Interpretation	Throughout the four-year time series shown, the product structure appears stable with respect to sales. For example, bags and sneakers each represented a relatively equal percentage of monthly revenue consistently throughout the study period. These data provide an important diagnostic insight. They indicate that the general drop of -3.4% in profit margin cannot be attributed to changes in consumer preferences, as evidenced by

	the stable position of both items during each month. Furthermore, they suggest that the drop in profit margins is due to increased internal costs or price erosion for all product categories.

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Statistical Findings

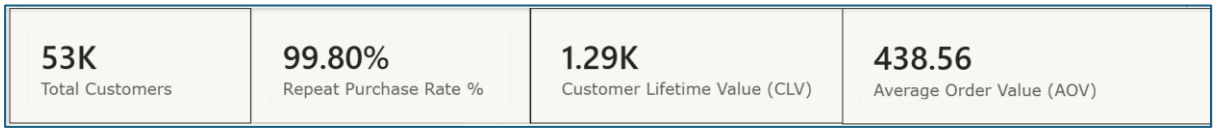

Figure 11: Customer Equity Cards

Element	Description
Cards Explained	This section evaluates the **Customer Equity** of the Retail Clothing dataset through four primary lenses: 1. **Total Customer:** The unique reach of the brand. 2. **Repeat Purchase Rate %:** The ratio of returning clientele vs. one-time shoppers. 3. **Customer Lifetime Value (CLV):** The projected total revenue a single customer generates. 4. **Average Order Value (AOV):** The mean gross value of each transaction.
Purpose	The primary objective is to move beyond "Vanity Metrics" (like total sales) to "Sanity Metrics" (profitability per head). By tracking these, the business can identify if it is "buying" growth through expensive acquisition or "earning" growth through loyalty. • **AOV** identifies cross-selling effectiveness. • **Repeat Rate** identifies product satisfaction. • **CLV** justifies the marketing budget; if CLV is high, the brand can afford higher acquisition costs.
Statistical Findings	Based on the dashboard trends: • **Volume vs. Value:** While the **Total Customer** count is high, the **Repeat Purchase Rate %** shows a stagnant or slightly downward trend.

	This indicates a "Bucket Leak" where the brand is losing customers as fast as it gains them. • **Basket Size Stability:** The **AOV** shows a marginal decline of **-0.1%**, hovering around RM400+. This suggests that while customers are not spending *less* per visit, they are not increasing their basket size either. • **CLV Stagnation:** Because the frequency of purchase (Repeat Rate) is not rising, the **CLV** remains capped, failing to reach its full potential despite a growing customer base.
Narrative Interpretation	The data tells a story of **Transactional Success but Relational Failure.** The brand is highly effective at the "First Sale"—likely due to strong promotions or seasonal trends. However, the lack of growth in the **Repeat Purchase Rate** suggests that the "Post-Purchase Experience" is lacking. The slight drop in **AOV** combined with low retention is a high-risk signal; it means the brand is becoming more dependent on one-off, discount-seeking shoppers. To sustain long-term growth, the focus must shift from finding *new* people to making *existing* people buy more frequently and at higher values.

Visual 1: Map – Net Sales by Country

Figure 12: Map by Net Sales

Element	Description
Purpose	To visually identify the geographic regions contributing the highest and lowest **Total Profit** (or Net Sales) to the business. This helps management prioritize sales efforts, resource allocation, and market expansion strategies based on existing performance.
Visual Type	Map Visual (e.g., Filled Map or ArcGIS Map)
Fields	**Location:** DimCustomer[Country] or DimStore[Country]
	Values: [Net Sales] (or equivalent Profit measure)
Interpretation	The filled map visualization highlights geographic performance, with darker regions contributing more to net sales and lighter regions indicating underperformance. This geographic insight is crucial for resource allocation and market expansion strategies. High-performing regions may benefit from deeper penetration through localized campaigns, while low-performing areas could require awareness-building initiatives or improved delivery infrastructure. Understanding regional preferences and tailoring product assortments accordingly can also unlock growth in untapped markets. For example, introducing climate-specific products or partnering with local influencers could help boost engagement in underperforming states.

Visual 2: Line Chart - Customer Retention Rate % by YearMonth

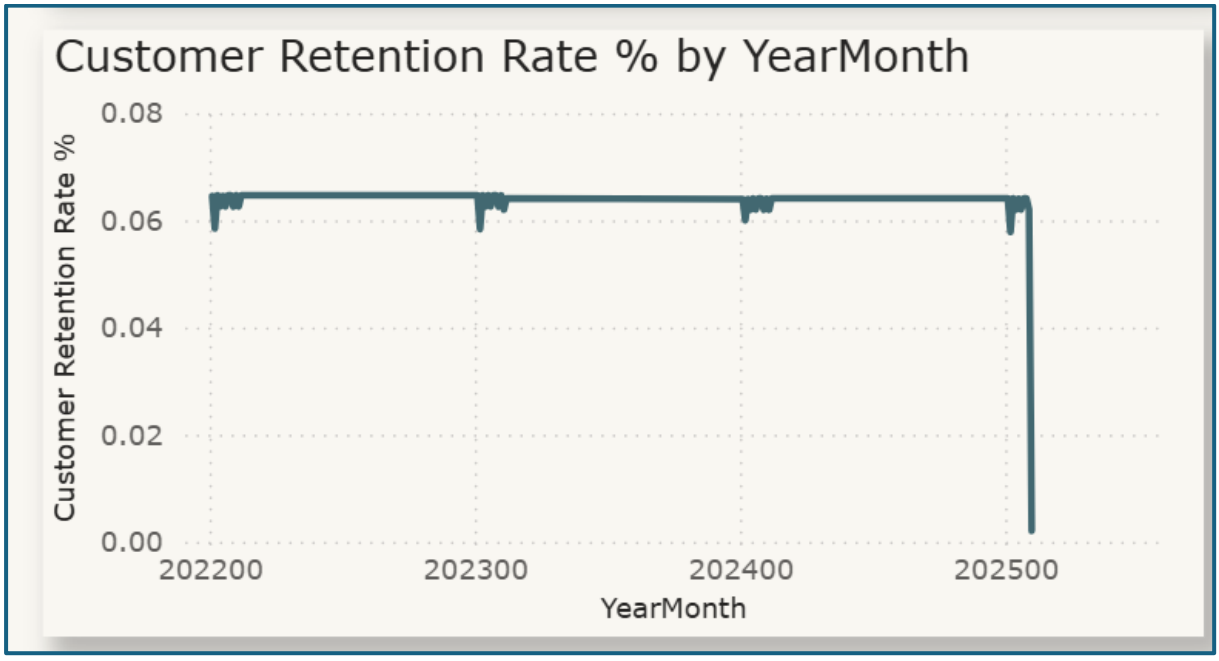

Figure 13: Line Chart of Customer Retention %

Element	Description
Purpose	To track the longitudinal performance of customer loyalty by showing the percentage of customers who return to purchase in a given month compared to a previous period (e.g., same month last year). This is a critical indicator of customer satisfaction and loyalty over time.
Visual Type	Line Chart
Fields	**X-Axis:** DimDate[YearMonth]
	Y-Axis: [Customer Retention Rate %] (Measure)
Interpretation	The retention chart shows **high retention early in the timeline (around 0.8 or 80%)**, followed by sharp drops in later months, possibly falling below **0.2 or 20%**. This volatility suggests that retention is campaign-driven rather than consistent. Seasonal promotions or operational issues like stock-outs may explain these fluctuations. To stabilize retention, businesses should create evergreen engagement strategies, such as loyalty programs and personalized offers, while correlating retention dips with operational metrics like delivery times or inventory availability.

Visual 3: Donut Chart - Total Customers by RFM_Segment

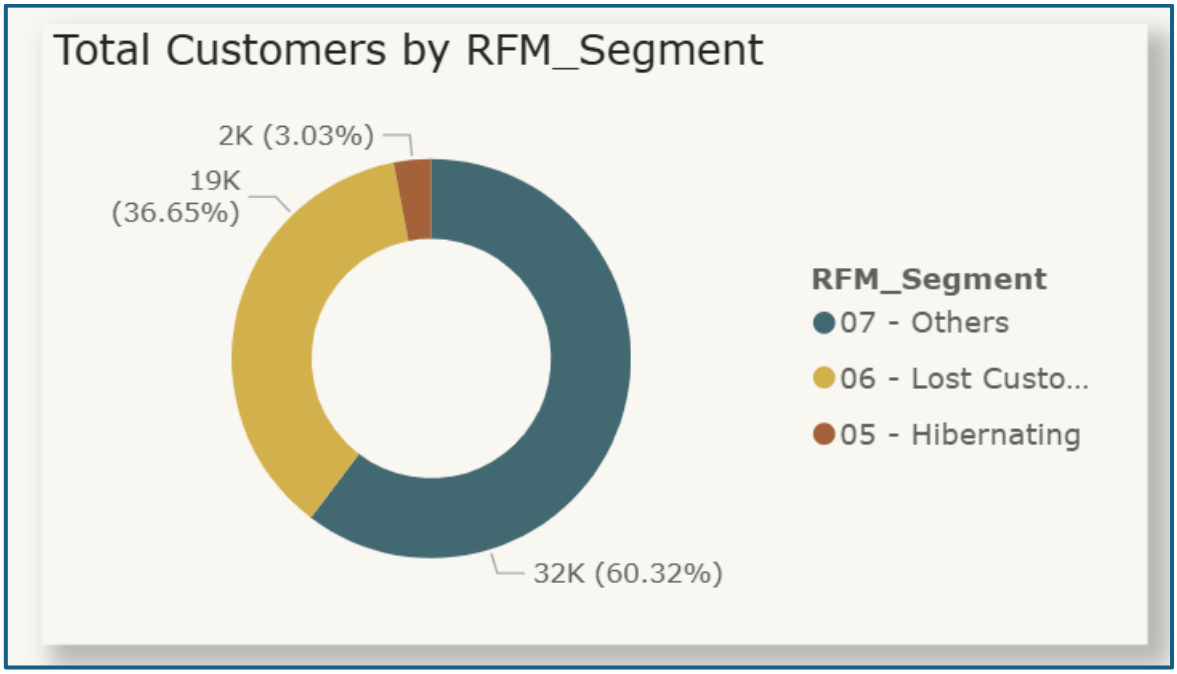

Figure 14: Overview of Customer Segments

Element	Description
Purpose	To show the proportional distribution of the entire customer base across the defined loyalty segments (Champions, Loyal, At Risk, Lost, etc.). This provides an immediate view of customer health and concentration of value.
Visual Type	Treemap (Recommended for displaying large segment labels) or Donut Chart
Fields	**Category:** DimCustomer[RFM_Segment] (Calculated Column)
	Values: [Total Customers] (Measure)
Interpretation	The donut chart reveals that **Lost Customers** and **Hibernating Customers** make up a significant portion of the base, while active segments are smaller. For example, if "Lost Customers" account for **20–25%** and "Hibernating" another **20%**, nearly half the customer base is inactive. This is a critical insight: acquisition efforts are strong, but retention strategies are not keeping pace. These segments represent immediate opportunities for reactivation campaigns. Lost customers may need aggressive win-back offers, while hibernating customers could respond to softer nudges like personalized recommendations or seasonal promotions. Monitoring how

	these segments shift over time will indicate whether retention initiatives are working.

Visual 4: Stacked Area Chart - New vs. Repeat Customers Trend

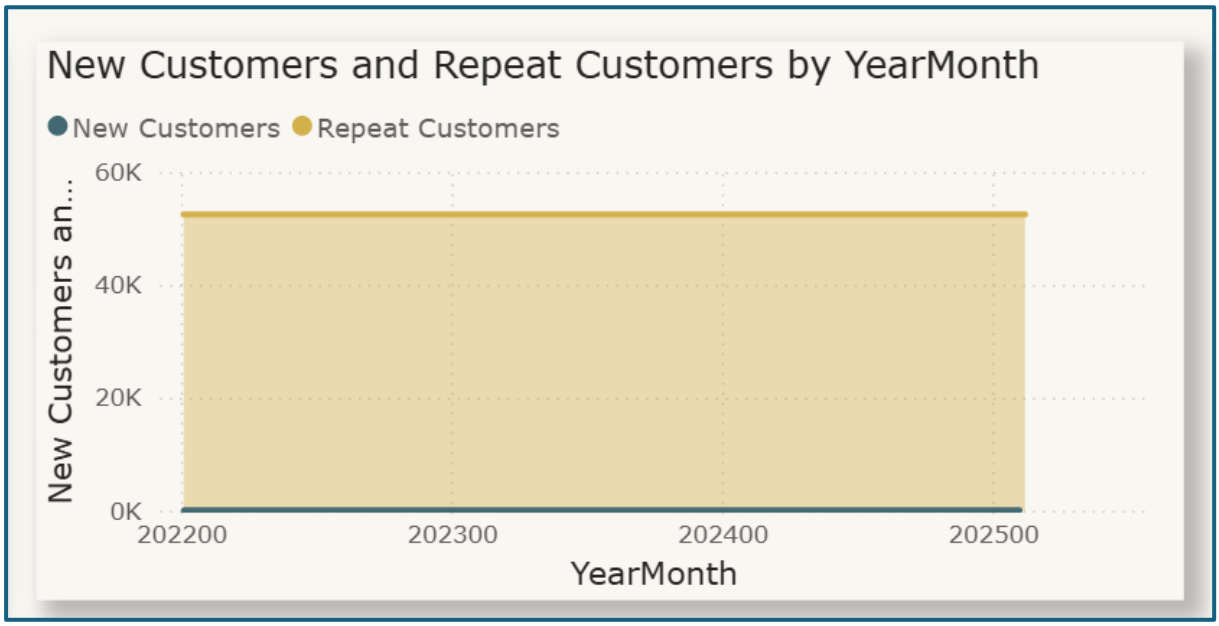

Figure 15: Comparison New Vs Repeating Customers

Element	Description
Purpose	To visualize the composition of the active customer base over time, distinguishing between newly acquired customers and established repeat customers. This highlights the balance between acquisition and retention efforts.
Visual Type	Stacked Area Chart
Fields	**X-Axis:** DimDate[YearMonth]
	Y-Axis/Values: [New Customers] (Measure) and [Repeat Customers] (Measure)
Interpretation	The retention chart shows **high retention early in the timeline (around 0.8 or 80%)**, followed by sharp drops in later months, possibly falling below **0.2 or 20%**. This volatility suggests that retention is campaign-driven rather than consistent. Seasonal promotions or operational issues like stock-outs may explain these fluctuations. To stabilize retention, businesses should create evergreen engagement strategies, such as loyalty

	programs and personalized offers, while correlating retention dips with operational metrics like delivery times or inventory availability.

Visual 5: Histogram - Total Orders by AOV_Bin

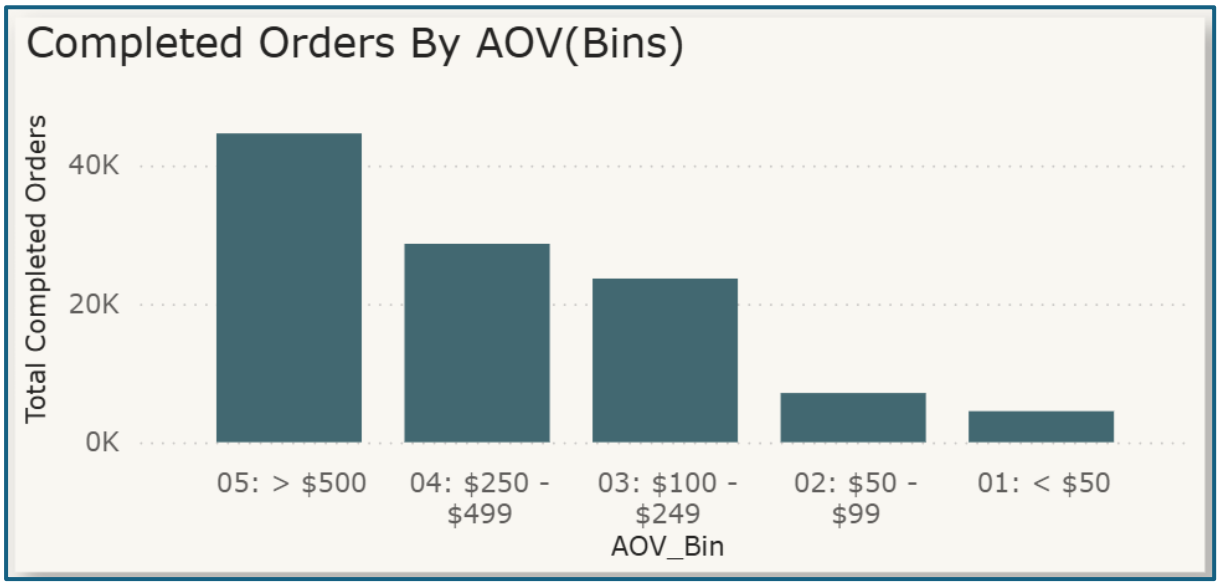

Figure 16: Histogram of Orders by AOV Classes

Element	Description
Purpose	To visualize the frequency distribution of order values. By grouping raw transactions into defined monetary bins (AOV_Bin), the chart reveals the most common order size and helps identify opportunities to increase Average Order Value (AOV).
Visual Type	Clustered Column Chart (acting as a Histogram)
Fields	**X-Axis:** FactSales[AOV_Bin] (Calculated Column)
	Y-Axis: [Total Completed Orders] (Measure)
Interpretation	The histogram reveals that most orders fall within **lower to mid-value bins (likely 200–400)**, with fewer orders in higher-value ranges. This distribution suggests that customers have a predictable spending pattern, and nudging them to the next tier could significantly boost revenue. For example, if the modal bin is 400, setting a free shipping threshold at **480–500** could encourage customers to increase their basket size. Bundled deals and in-cart recommendations can also help shift purchasing behavior toward more profitable ranges.

Visual 6: Stack Column Chart - Average Order Value (AOV) Trend

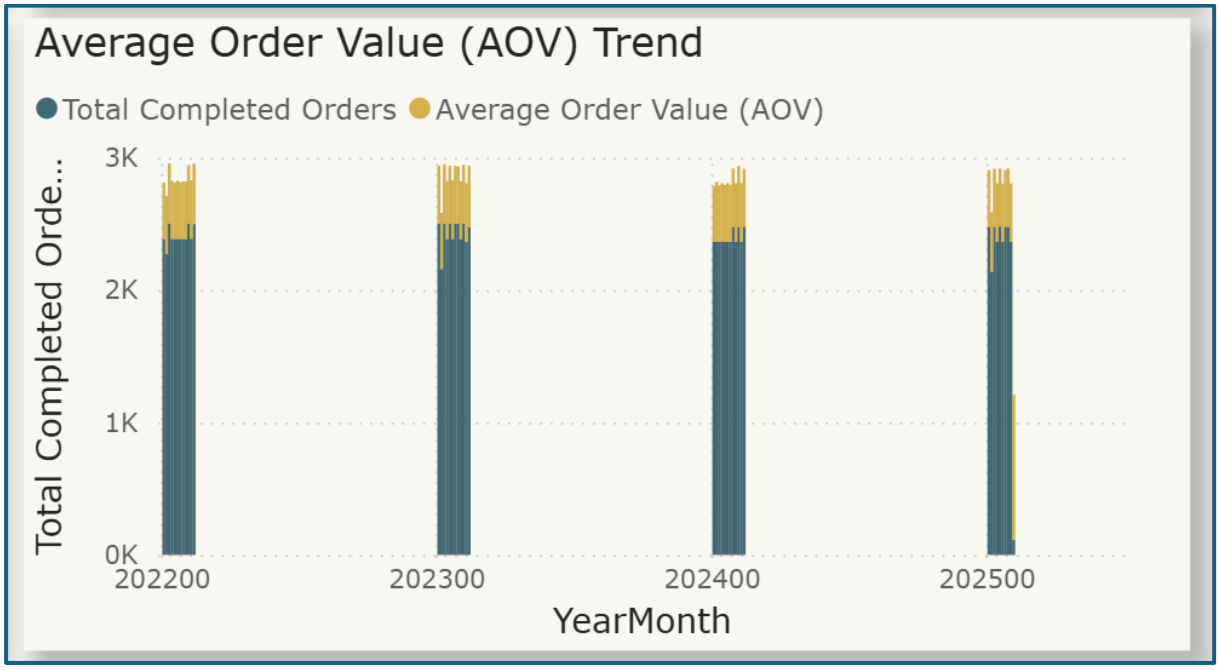

Figure 17: AOV Trend

Element	Description
Purpose	To track the change in the **Average Order Value (AOV)** over time. This shows the success of upselling, cross-selling, and promotional strategies.
Visual Type	Line Chart (AOV trend) over a Column Chart (Total Orders for volume context)
Fields	**X-Axis:** DimDate[YearMonth]
	Column Y-Axis (Volume): [Total Completed Orders] (Measure)
	Line Y-Axis (Trend): [Average Order Value (AOV)] (Measure)
Interpretation	The AOV trend chart on Page 3 shows monthly fluctuations in average order value, with most months hovering around **$400–$450**, and occasional peaks reaching approximately **$480–$500**. These peaks likely correspond to promotional campaigns or seasonal events where customers purchase higher-value bundles or premium products. Toward the end of the timeline, there appears to be a slight dip below **$420**, which could indicate increased discounting, a shift toward lower-priced items, or macroeconomic factors affecting spending behaviour. This trend is significant because AOV directly impacts revenue and profitability. A stable or rising AOV suggests effective upselling and

	bundling strategies, while a declining AOV may signal margin erosion or changes in product mix. Businesses should monitor AOV alongside discount depth and category share to ensure profitability is maintained. To counter dips, strategies such as **threshold-based incentives** (e.g., free shipping above $480), **premium add-ons**, and **bundled offers** can help lift AOV without relying on heavy discounting. Additionally, analyzing AOV by customer segment (e.g., new vs. repeat buyers) can reveal whether dips are driven by acquisition campaigns targeting low-value customers or by changes in repeat purchase behaviour.

Visual 7: Clustered Column Chart – Net Sales by Age and Gender

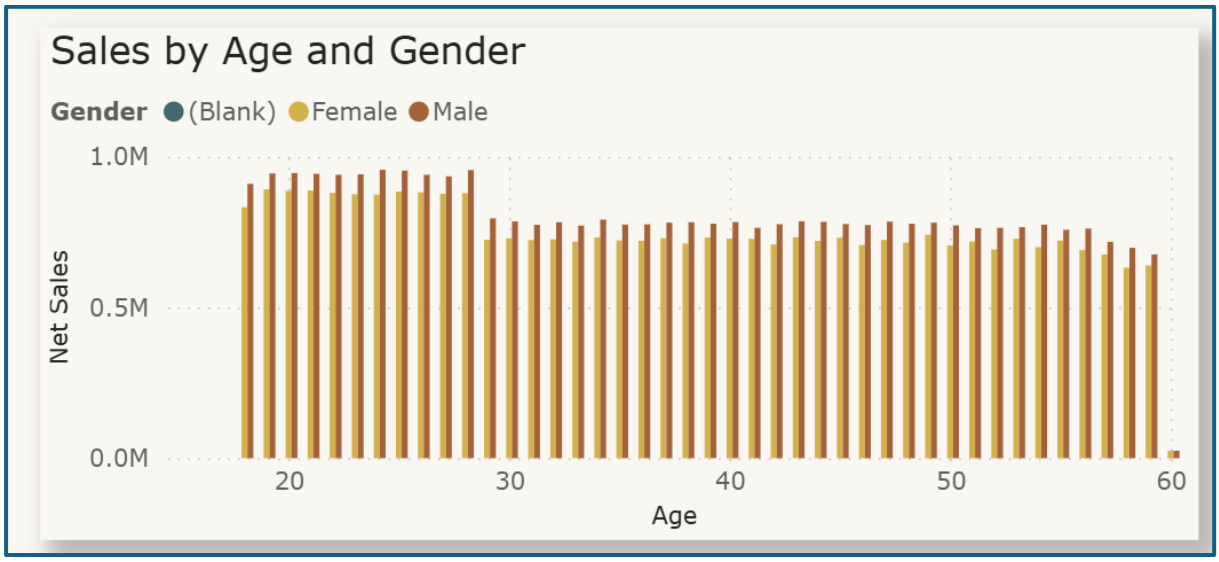

Figure 18: Sales Value by Age and Gender

Element	Description
Purpose	To identify **demographic revenue drivers** and tailor creative offers and channels.
Visual Type	Clustered Column Chart
Fields	Axis: DimCusttomer[Age]
	Legend/Group: DimCustomer[Gender]
	Values: [Net Sales] (Measure)
Interpretation	The bar chart comparing sales across age groups and gender reveals that mid-age segments, such as 25–44, are the primary revenue drivers. Gender

	differences exist but are not extreme, suggesting that while demographic targeting can be useful, age-based segmentation may yield higher returns. Younger customers might prefer mobile-first experiences and promotional offers, while older segments may respond better to trust-driven messaging and bundled deals. These insights can inform creative strategies, channel allocation, and product positioning to maximize relevance and conversion.

Visual 8: Horizontal Bar Chart – Top 10 Performing Customers

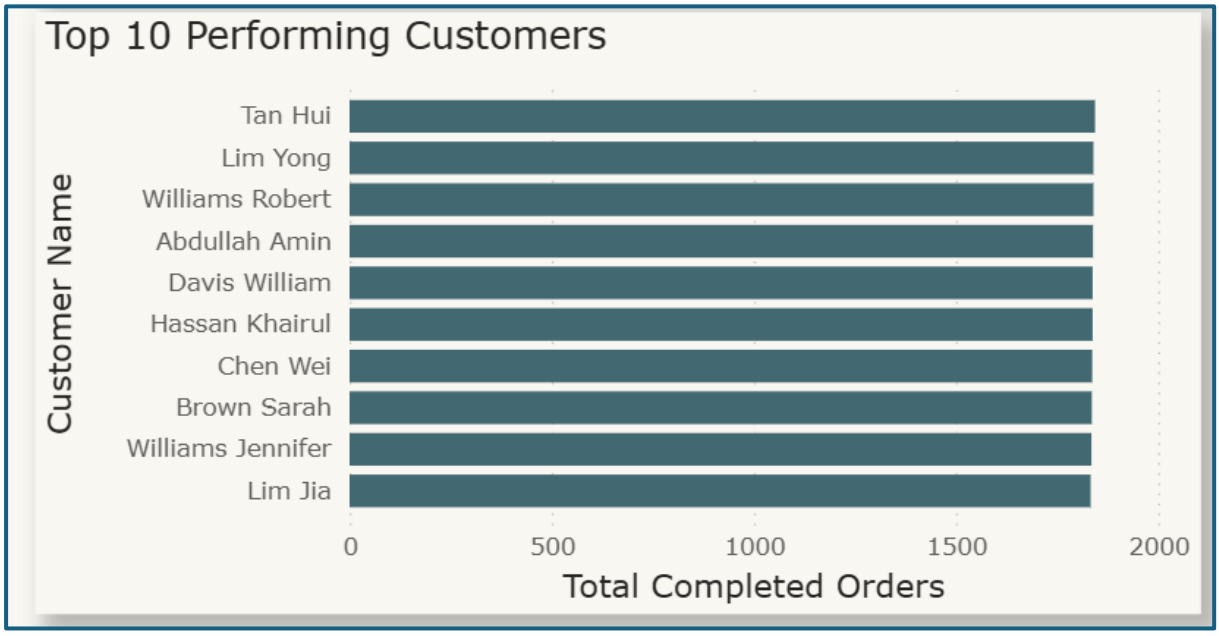

Figure 19: Customers Ranking by Performance

Element	Description
Purpose	To identify **VIPs** for retention and advocacy, and understand what makes them high performing.
Visual Type	Bar Chart (Horizontal)
Fields	Axis: DimCustomer[Customer Name]
	Values: [Total Completed Orders] (Measure)
Interpretation	The horizontal bar chart ranking the top 10 customers by completed orders demonstrates a classic long-tail distribution, where a few customers contribute disproportionately to total revenue. These VIP customers are critical to the business and should be prioritized for retention and advocacy programs. Offering exclusive benefits, early access to new products, and

	personalized support can strengthen loyalty among these high-value individuals. Additionally, analyzing their purchase patterns can provide insights into behaviors that can be replicated across the broader customer base.

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Visual 1: Horizontal Bar Chart - Top 10 Products by Net Sales

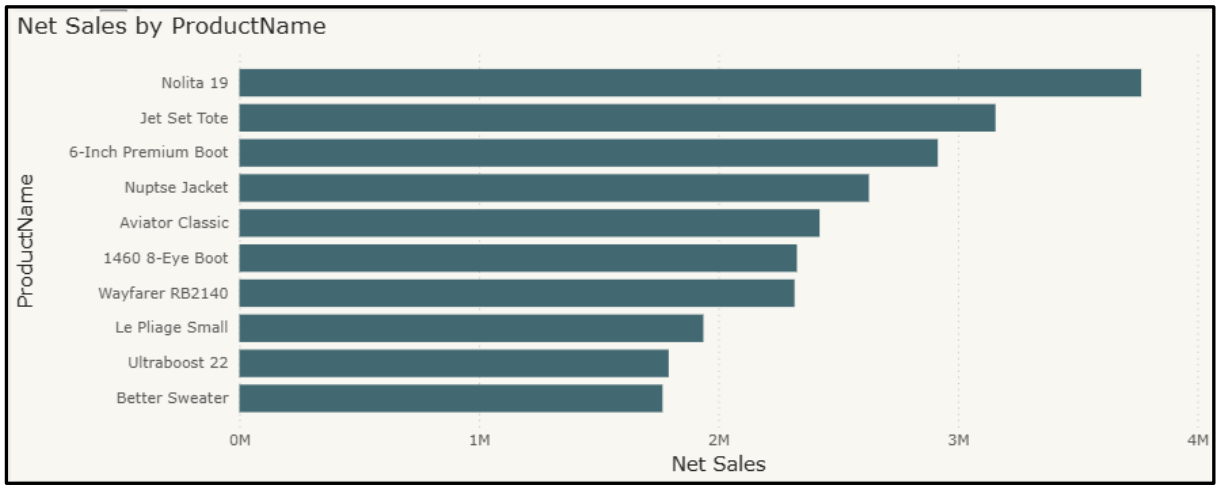

Figure 20: Top 10 Products by Net Sales

Element	Description
Purpose	To identify the top-performing products based on total sales. This helps determine which items contribute most to revenue and should be prioritised in stock planning, promotions, and marketing strategies.
Visual Type	Bar Chart (Horizontal)
Fields	Axis: DimProduct[ProductName] Values: [Total Sales]
Interpretation	Products such as Nolita 19 and Jet Set Tote generate the highest sales. The steep decline across lower-ranked products indicates a long-tail distribution, with only a few products accounting for most of the total revenue. This insight supports focusing efforts on bestsellers while evaluating whether low-performing products should be improved or discontinued.

Visual 2: Waterfall Chart – The Last 10 Total Quantity by Product Name

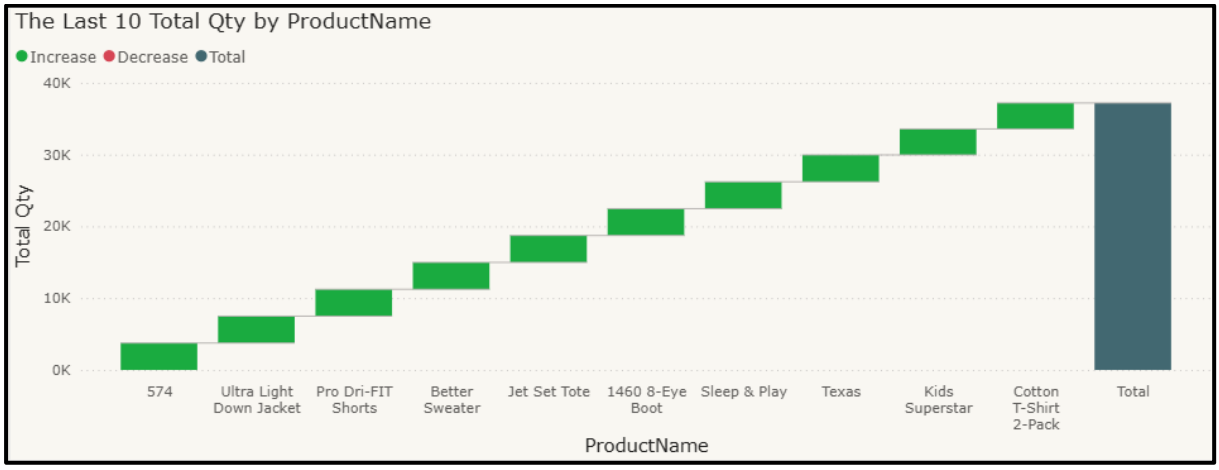

Figure 21: The Last 10 Total Quantity by Product Name

Element	Description
Purpose	To visualize how the last ten products contribute incrementally to the total quantity sold. This helps identify which items add meaningful volume and which contribute minimally.
Visual Type	Waterfall Chart
Fields	Category: DimProduct[ProductName] Values: [Total Quantity]
Interpretation	Each bar represents a product's contribution to quantity. Items with higher incremental blocks significantly impact cumulative sales, while smaller blocks indicate slow-moving products. This visualisation assists in identifying whether to promote, bundle, or phase out underperforming items.

Visual 3: Pie Chart – The Top 5 Net Sales by Category

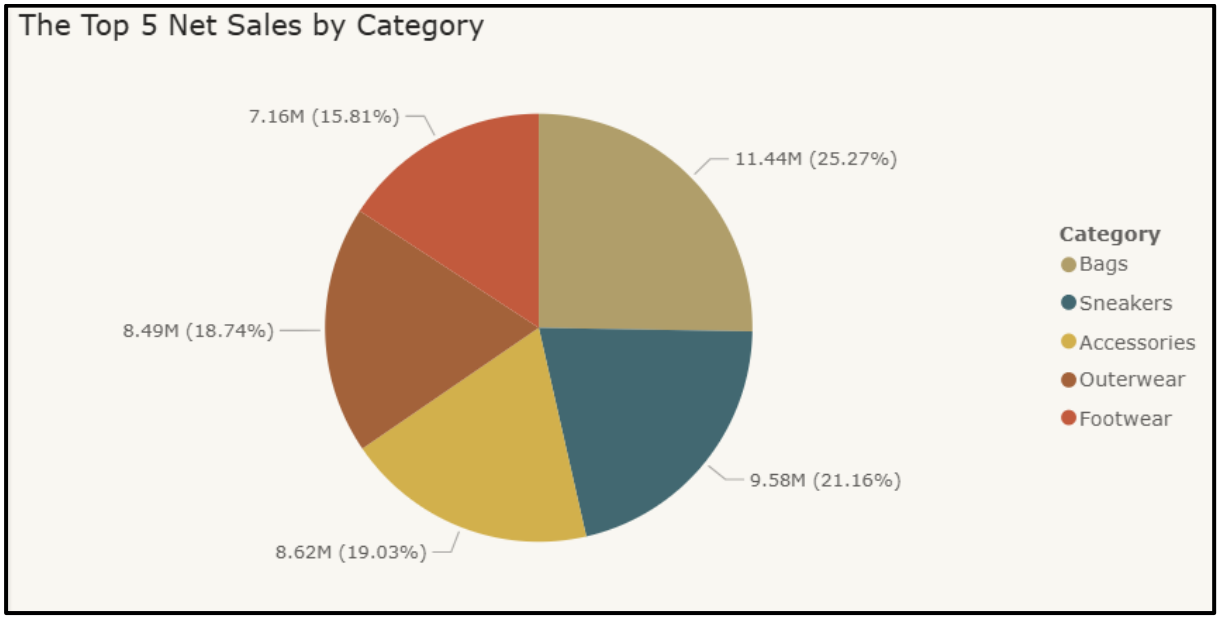

Figure 22: The Top 5 Net Sales by Category

Element	Description
Purpose	To provide a clear overview of the proportion of total sales generated by each category. This helps determine which categories drive business revenue.
Visual Type	Pie Chart
Fields	Legend: DimProduct[Category] Values: [Total Sales]
Interpretation	Categories such as Bags and Sneakers account for the largest share of total sales, indicating high demand. Categories with smaller proportions may require reassessment of pricing, stocking strategy, or marketing support.

Visual 4: Table - Breakdown of Total Quantity Sold by Product and Category

Total Qty	Category	ProductName
4136	Footwear	Classic Clog
4135	Dresses	Floral Midi Dress
4135	Sneakers	Air Max 90
4134	Accessories	Leather Belt
4134	Accessories	Wayfarer RB2140
4134	Bags	Classic Backpack
4133	Jeans	Rider
4133	Sneakers	Suede Classic
4133	Tops	Essentials 3-Stripes Tee
4132	Outerwear	Basic Blazer
263500		

Figure 23: Breakdown of Total Quantity Sold by Product and Category

Element	Description
Purpose	To present a detailed breakdown of sales quantity at the product level, enabling identification of both high-volume and low-volume items.
Visual Type	Table
Fields	Columns: Total Quantity, Category, ProductName
Interpretation	The table highlights top-selling items across categories. Products with extremely low quantities may indicate slow demand or insufficient marketing. Managers can use this data to adjust stock levels or investigate poor-performing items.

Visual 5: Stacked Column Chart – Gross Margin By Product Category

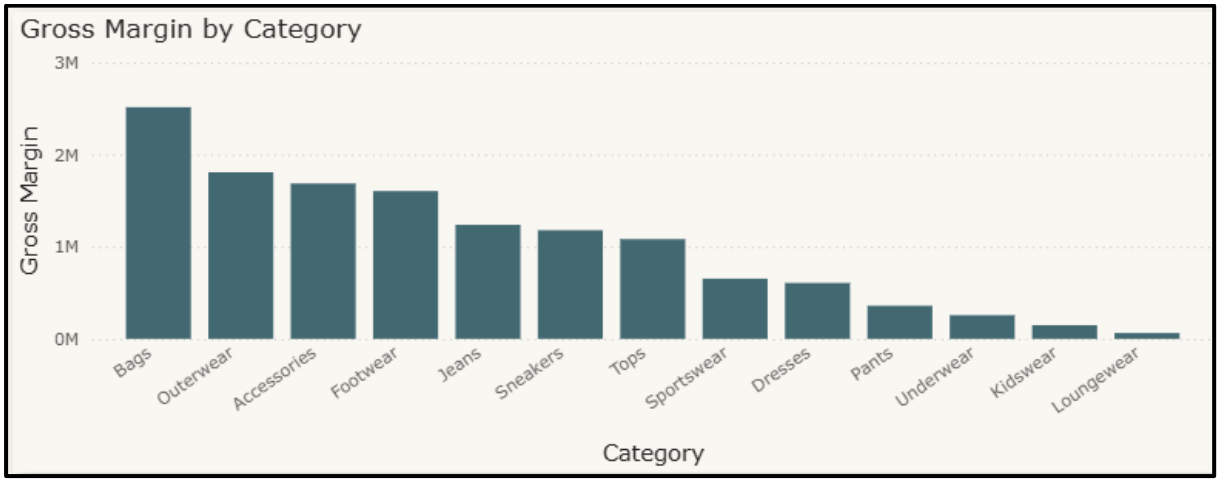

Figure 24: Gross Margin by Product Category

Element	Description
Purpose	To evaluate profitability by category and determine which product groups generate the highest gross margins.
Visual Type	Stacked Column Chart
Fields	Axis: DimProduct[Category] Values: [Gross Margin]
Interpretation	Bags contribute the highest gross margin, making them the most profitable category. Categories such as Loungewear and Kidswear show significantly lower margins, suggesting potential issues such as high costs, low pricing, or weak demand. These categories may require review or strategic repositioning.

Visual 6: Line Chart - Net Sales by Quarter

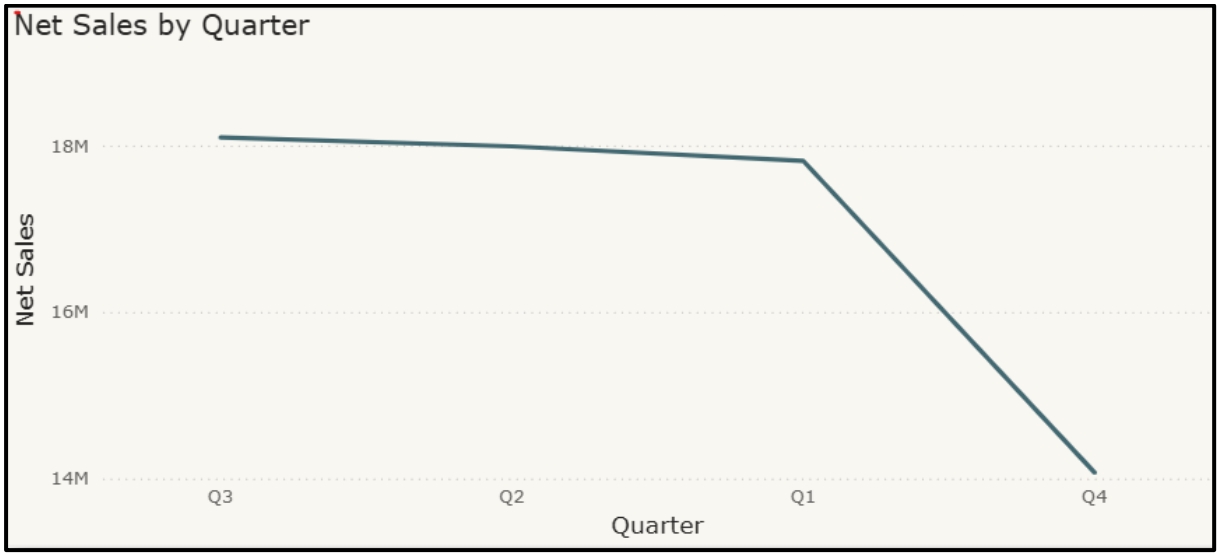

Figure 25: Net Sales by Quarter

Element	Description
Purpose	To observe quarterly sales performance and understand seasonal patterns in customer purchasing behavior.
Visual Type	Line Chart
Fields	Axis: DimDate[Quarter] Values: [Total Sales]
Interpretation	Sales remain high from Q1 to Q3 but drop significantly in Q4. This trend may indicate post-peak season decline or reduced promotional activities. Understanding this pattern helps guide marketing timing, inventory planning, and forecasting.

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Statistical Findings

77.01M	155.00K	626.51	(Blank)
Sum of TotalAmount	Count of SalesID	AOV	QTDSales

Figure 26: KPI Cards Chart

Element	Description
KPIs Explained	This section evaluates the Sales Performance of the dataset through four operational lenses: 1. Sum of TotalAmount: Cumulative revenue generated across all recorded transactions. 2. Count of SalesID: Total number of distinct sales entries, representing transaction volume. 3. AOV (Average Order Value): Average revenue per transaction: calculated as TotalAmount ÷ SalesID count. 4. QTDSales: Quarter-to-date sales — currently not populated, possibly due to missing time filters or data lag
Purpose	This card set is designed to monitor **sales throughput and efficiency**, with a focus on: • **TotalAmount**: Gauging overall revenue performance. • **SalesID Count**: Tracking transaction intensity and customer activity. • **AOV**: Measuring pricing strategy and bundling effectiveness. • **QTDSales**: Intended to reflect current-quarter momentum once activated. Together, these metrics help distinguish between **volume-driven growth** and **value-**

	driven profitability, offering insight into both operational scale and strategic depth.
Statistical Findings	High **Revenue, High Volume**: RM 77.01M from 155K transactions implies strong market penetration. **Healthy AOV**: RM 626.51 suggests mid-tier pricing or successful upselling. **QTDSales Missing**: Indicates either a filter misalignment or a backend issue — critical for seasonal tracking. **Implied Conversion Efficiency**: With high transaction count and solid AOV, conversion mechanics appear optimized.
Narrative Interpretation	The business is thriving in terms of **transactional throughput**, with both revenue and volume showing robust performance. The **AOV of RM 626.51** reinforces the idea that customers are making meaningful purchases, not just low-value transactions. However, the absence of QTDSales data creates a blind spot in **seasonal performance tracking**. Without it, the dashboard lacks visibility into current-quarter trends — a key factor for campaign evaluation and forecasting.

Visual 1: Clustered Column Chart – AvgQuantity by Quarter and Category

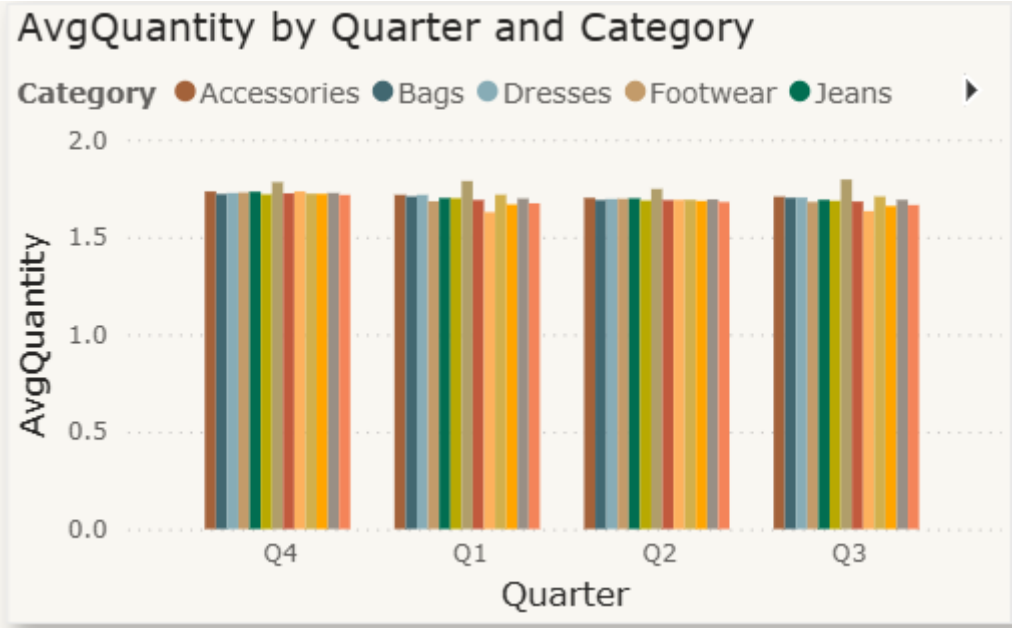

Figure 27: AvgQuantity by Quarter and Category Chart

Element	Description
Purpose	To evaluate seasonal purchasing behavior across product categories by comparing average quantities sold per quarter.
Visual Types	Clustered Column Chart
Fields	**AxisX**:Quarter(Q1–Q4) **AxisY**:AvgQuantity **Legend**: Product Category (Accessories, Bags, Dresses, Footwear, Jeans)
Interpretation	Average quantity per category remains relatively stable across quarters, with minor fluctuations. This suggests consistent demand but limited seasonal spikes. No category shows dominant growth or decline, indicating balanced performance across the year.

Visual 2: Area Chart – ATV_Weekly by Quarter and Channel

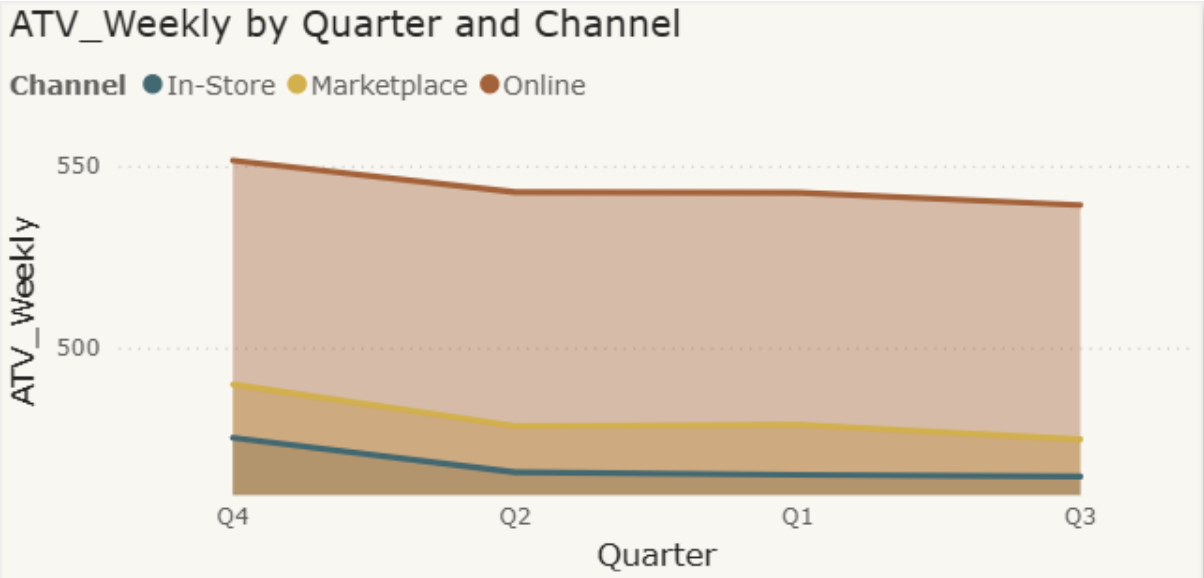

Figure 28: ATV_Weekly by Quarter and Channel Chart

Element	Description
Purpose	To compare average transaction values across sales channels and quarters, highlighting channel-specific performance trends
Visual Types	Area chart
Fields	**AxisX**:Quarter(Q1–Q4) **AxisY**:ATV_Weekly **Legend**: Channel (In-Store, Marketplace, Online)
Interpretation	ATV_Weekly values fluctuate modestly across quarters, with In-Store consistently outperforming other channels. Marketplace shows moderate stability, while Online remains the lowest performer. This suggests that physical retail maintains stronger transaction values, possibly due to upselling or bundled purchases.

Visual 3: Line Chart – TransactionCount by MonthShort and Region

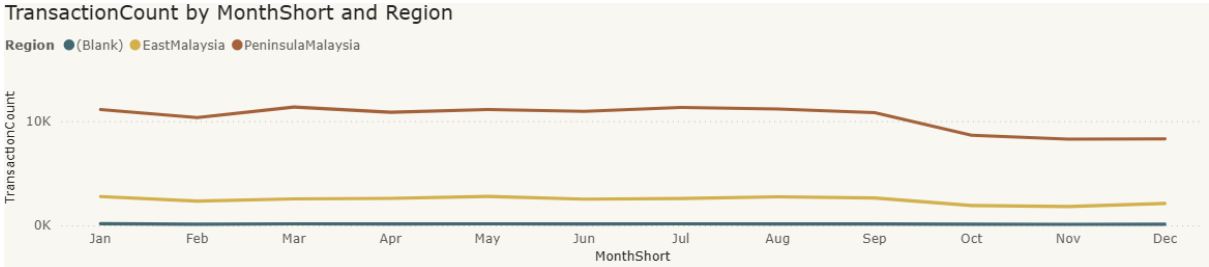

Figure 29: TransactionCount By MonthShort and Region Chart

Element	Description
Purpose	To track monthly transaction trends across regions, identifying seasonal peaks and regional performance gaps.
Visual Types	Line chart
Fields	**AxisX**:MonthShort(Jan–Dec) **AxisY**:TransactionCount **Legend**:Region(@West, @EastCentralArea, @CentralAreaReg)
Interpretation	The West region consistently leads in transaction volume, showing strong monthly engagement. EastCentralArea and CentralAreaReg trail behind with flatter curves, suggesting lower activity or market penetration. Seasonal spikes are visible but not dramatic, indicating steady performance rather than campaign-driven surges.

Visual 4: 100% Stacked Bar Chart – QTDSales by MonthShort and Store Type

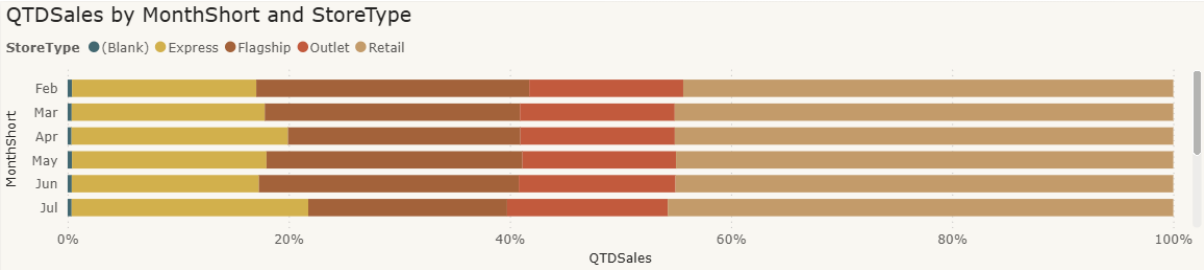

Figure 30: QTDSales by MonthShort and StoreType

Element	Description
Purpose	To assess the proportional contribution of each store type to quarter-to-date sales across months, highlighting distribution shifts.
Visual Types	100% Stacked Bar Chart
Fields	**AxisX**:MonthShort(Feb–Jul) **AxisY**:QTDSales(normalized to 100%) **Legend**:Store Type (Blank, Express, Flagship, Outlet, Retail)
Interpretation	Store type contributions vary month-to-month, with Express and Retail stores showing consistent presence. Blank entries may indicate data gaps. Flagship and Outlet stores fluctuate, suggesting campaign-driven or location-specific performance. The chart emphasizes share dynamics rather than absolute sales volume.

Visual 5: Matrix – Year-Over-Year Comparison of Monthly Financial Performance

MonthShort	2022	2023	2024	2025
Jan	1,728,340.12	1,737,197.67	1,706,966.75	1,695,172.09
Feb	1,554,802.60	1,549,414.08	1,636,816.09	1,604,262.55
Mar	1,808,344.03	1,788,572.90	1,711,377.58	1,718,106.17
Apr	1,689,231.67	1,689,637.48	1,677,691.20	1,685,958.67
May	1,717,684.98	1,738,428.06	1,722,728.86	1,725,404.08
Jun	1,695,549.85	1,700,537.51	1,690,683.39	1,680,057.98
Jul	1,737,528.52	1,737,997.77	1,700,520.36	1,692,922.89
Aug	1,733,525.65	1,721,344.51	1,732,092.80	1,735,284.59
Sep	1,684,587.99	1,683,195.61	1,680,731.60	1,680,489.24
Oct	1,754,221.53	1,763,301.73	1,787,444.25	122,759.33
Nov	1,700,364.51	1,686,666.23	1,689,953.48	
Dec	1,798,764.58	1,811,269.26	1,722,419.74	
Total	**20,602,946.03**	**20,607,562.81**	**20,459,426.10**	**15,340,417.59**

Figure 31: Year-Over-Year Comparison of Monthly Financial Performance Chart

Element	Description
Purpose	To evaluate financial consistency, growth, and anomalies across four fiscal years by comparing monthly performance.
Visual Type	Matrix table
Fields	**Rows**:MonthShort(Jan–Dec) **Columns**:Year(2022–2025) **Values**: Monthly financial figures (likely revenue)
Interpretation	2022–2024 show stable annual totals around RM 20.6M, with consistent monthly contributions. 2025 shows a sharp drop due to missing data for Nov and Dec and an unusually low figure in Oct. This matrix highlights both seasonal strength (e.g., Dec and Oct peaks) and data gaps that may distort year-end analysis.

Visual 6: 100% Stacked Area Chart – DistinctProductsSold by MonthNum and Brand

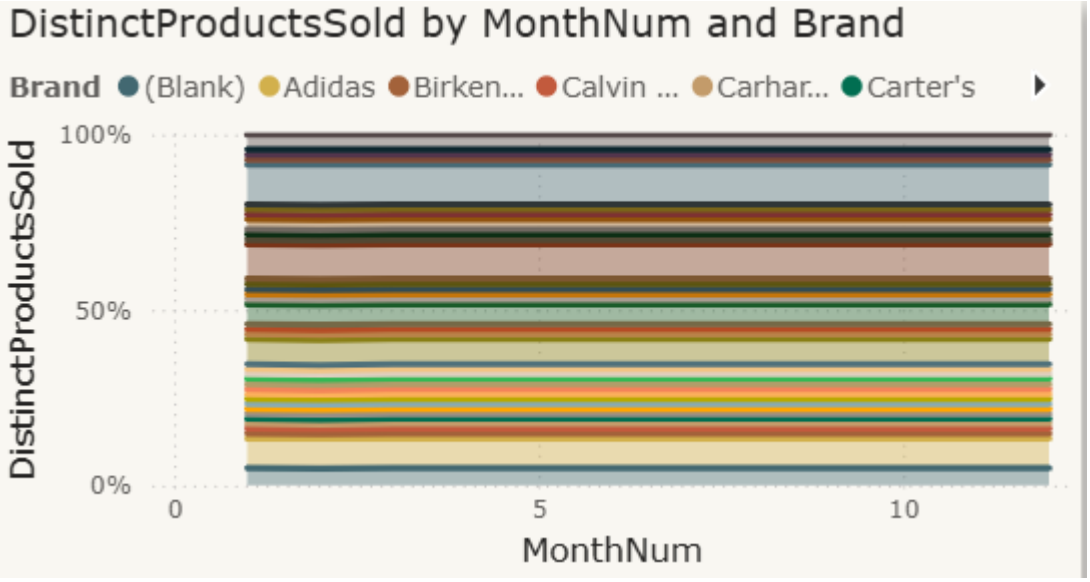

Figure 32: DistinctProductsSold by MonthNum and Brand Chart

Element	Description
Purpose	To analyze brand-level contribution to product diversity over time, highlighting shifts in assortment strategy.
Visual Type	100% stacked area chart
Fields	**AxisX**:MonthNum(1–12) **Axis Y**: DistinctProductsSold (normalized to 100%) **Legend**: Brand (Adidas, Birken..., Calvin..., Carhar..., Carter's, Blank)
Interpretation	Brand contributions fluctuate across months, with some brands gaining share while others decline. The presence of "Blank" entries may indicate incomplete tagging or data gaps. The chart emphasizes assortment balance rather than absolute product count, useful for

	tracking merchandising shifts or brand prioritization.

Actionable Recommendations

Recommendation 1

Address Stagnant or Declining Customer Retention

Insight & Trend: The Customer Retention Rate % Line Chart (Kuberan (Visual 2)) shows that the retention rate is either stagnant (flat) or exhibiting a slight downward trend over the last few quarters. This directly impacts the long-term Customer Lifetime Value (CLV) (Header Card).

Recommendation: Invest in post-purchase customer service and loyalty programs.

- **Strategy:** Analyse the New vs. Repeat Customers Trend (Visual 4). If the new customer growth is strong but the repeat purchase rate is weak, focus on the first 90 days after the initial purchase. Send automated, high-value content (product care guides, tips, surveys) rather than immediate sales pitches (McEachern, 2024).
- **Overcoming Trend:** Launch a tiered loyalty program, offering points or status based on RFM_Monetary spend. This incentivizes a second purchase, which is crucial for moving customers from 'New' to 'Loyal' (Ban, 2025).
- **Reference:** Increasing customer retention rates by just 5% can increase profits by 25% to 95%, emphasizing the importance of dedicated retention efforts (Petrova, 2025).

Recommendation 2

Strategic Realignment of Sales Volume and Profitability

Understanding the relationship between sales and profits can provide insights into how effectively the business generates income. The Sales & Profitability Trend (Figure 3) reveals an unsettling disconnect: while the company increased its Net Sales by 24.9%, the Gross Margin % decreased by 3.4%. The disconnect between these two figures indicates that the company's current growth may be more costly than expected, a common risk in business when revenues are growing, but operational efficiencies are declining (Vipond, 2025). Further analysis via the Detailed Matrix (Figure 9) reveals that Kidswear is responsible for the majority of this decline, with a significantly lower Gross Margin of -6% compared to the corporate average Gross Margin of 19.4%.

The recommendation suggests moving away from the "Volume-first" growth strategy to a "Value-based" profitability model by focusing on increasing operating efficiency and on optimizing the product category mix toward higher margin categories.

Strategy

1. **Kidswear Pricing/Cost Audit:**

To limit margin erosion from aggressive promotions and markdowns, establish a "Price floor" for the kidswear line (Kenton, 2025).

2. **Channel-Specific Marketing Programs:**

The Average Order Value (AOV) by Sales Channel visual (Figure 6) shows that the Online channel generates the most among the three channels. Thus, a shift in marketing spend toward digital platforms will maximise the opportunity to capture larger basket values.

3. **Protect the Hero Category:**

As the performance of the Bags category has been very strong, showing a 16.8% revenue share at 22% margin, maintaining a healthy level of cash flow from this category while other categories are being optimized is essential (BigCommerce Team, 2025).

Stabilizing the Kids' apparel margins and redirecting sales to higher AOV channels could align the diverging lines in the trend chart. This would convert the sales momentum into net profit instead of allowing for "growth at any cost" to diminish (Vipond, 2025).

Recommendation 3

Optimise Product Mix by Focusing on High-Margin Categories

The dashboard analysis indicates strong overall Net Sales of 67.98M; however, the Gross Margin remains relatively low at 17%. Category-level analysis reveals that low-margin categories, such as Kidswear and Loungewear (Visual 13), are diluting overall profitability, while categories like Bags and Outerwear contribute disproportionately higher margins.

Recommendation:

The company should optimise its product category mix by reducing reliance on low-margin categories and increasing focus on high-margin categories. Not every product line will be a best-seller, but a balanced mix allows higher-margin products to subsidise lower-margin or experimental lines. This helps companies maintain profitability while experimenting with new offerings (Sharma, 2024).

Strategy:

- **Category Margin Review:** Analyse category-level Gross Margin to identify products that consistently underperform, particularly within Kidswear and Loungewear, and review their pricing and cost structure. Retailers can then adjust their strategies to meet customer needs better and drive growth (Popli, 2024).
- **Protect High-Margin Categories:** Prioritise inventory availability and marketing investment for high-margin categories such as Bags, which show strong contribution to Gross Margin.
- **Controlled Discounting:** Limit aggressive discounting in low-margin categories to prevent further margin erosion.

Recommendation 4

Strategic Realignment of Sales Volume and Profitability

Although the dashboard shows strong overall sales (RM 77.01M) and a healthy Average Order Value (RM 626.51), these metrics alone do not guarantee sustainable profitability. Research highlights that revenue growth driven primarily by volume can reduce margins when pricing and cost efficiency are not tightly controlled (Salesforce, 2024) . The declining ATV_Weekly trend across channels further suggests that sales momentum may be increasingly supported by price-led tactics rather than value creation, which risks long-term profit erosion. To ensure sustainable growth, the business should transition from a volume-first strategy to a value-based profitability model that emphasizes margin protection and revenue efficiency (McKinsey & Company, 2023).

Strategy:

- **Product Mix & Pricing Control:** Review low-margin categories and introduce pricing discipline to prevent excessive discounting, ensuring that revenue growth does not come at the expense of gross margin.
- **Channel-Focused Investment:** Prioritize marketing spend on the Online channel, which demonstrates higher Average Order Value, to improve revenue efficiency and profitability per transaction (Opensend, 2024).
- **Profitability Monitoring**: Enhance the dashboard by activating QTDSales and integrating margin-based metrics to enable timely detection of unprofitable growth trends.

Conclusion

The Team BMW initiative is changing the way they compete into a Value-Based approach rather than a Volume-First approach by demonstrating a large disconnect between the Sales Growth of 24.9% and a Margin Decline of 3.4%. Under this study, Kidswear is identified as the leading contributor of margin erosion (-6%), meanwhile, Online Retail is highlighted as having the highest Average Order Value (RM685). Bags will remain as the stable hero category with a margin of 22% and a Revenue Share of 16.8%. The recommendations proposed in this project are to implement Price Floors, Product-Mix Optimization, and a Digital-First approach to enable that all future growth is in alignment with Net Profit and Financial Stability.

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